My First LaTeX Document

Your Name

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Contents

1 Introduction **2**

2 Background **2**
2.1 Mathematical Expressions 2
2.2 Numbered List 2
2.3 Theorem 2

3 A Warning or Two **2**

4 Science! **3**
4.1 Schrödinger Equation 3

1 Introduction

This is my first LaTeX document.

Words are separated by one or more spaces. Paragraphs are separated by one or more blank lines.

2 Background

In Section 4, we discussed the introduction.

Double quotes are typed like this: “quoted text”.

Single quotes are typed like this: ‘single-quoted text’.

Long dashes are typed as three dash characters—like this.

Italic text is typed like this: *This is italic text.*

Bold text is typed like this: **This is bold text**.

2.1 Mathematical Expressions

$\beta = a^2 + b \times c + 2$

An example of a mathematical expression is $a^2 + b^2 = c^2$.

2.2 Numbered List

1. First item
2. Second item
3. Third item

2.3 Theorem

Theorem 2.1. *There are infinitely many prime numbers.*

As stated in Theorem **??**, there are infinitely many prime numbers.

3 A Warning or Two

If you get too much space after a mid-sentence period—abbreviations like etc. are the common culprits—then type a backslash followed by a space after the period.

Remember, special characters such as dollar sign and backslash must be used carefully in LaTeX.

The following characters are printed by typing a backslash in front of them:

\$ \& \# \% _ \{ \}

4 Science!

Einstein's famous mass-energy equivalence equation is:

$E = mc^2$

This equation describes the relationship between mass and energy.

Einstein introduced this relationship in his 1905 research paper [*Einstein*, 1905].

4.1 Schrödinger Equation

The time-dependent Schrödinger equation is a fundamental equation of quantum mechanics.

It is given by:

$$i\hbar\frac{\partial\Psi(\mathbf{r},t)}{\partial t} = \left[-\frac{\hbar^2}{2\mu}\nabla^2 + V(\mathbf{r},t)\right]\Psi(\mathbf{r},t) \tag{1}$$

As shown in Equation 1, the Schrödinger equation describes the evolution of the quantum wave function over time.

Schrödinger introduced this equation in his 1926 research paper [*Schrödinger*, 1926].

References

Einstein, A., Ist die trägheit eines körpers von seinem energieinhalt abhängig?, *Annalen der Physik*, *323*(13), 639–641, 1905.

Schrödinger, E., Quantisierung als eigenwertproblem, *Annalen der Physik*, *384*(4), 361–376, 1926.